

RATIO & PROPORTION

- If $\frac{a}{b} > 1$ then $\frac{a}{b} > \frac{a+x}{b+x}$, $(x > 0)$.
- If $\frac{a}{b} < 1$ then $\frac{a}{b} < \frac{a+x}{b+x}$, $(x > 0)$.
- If $\frac{a_1}{b_1}, \frac{a_2}{b_2}, \frac{a_3}{b_3}, \dots, \frac{a_n}{b_n}$ are unequal fractions, of which the denominators are of the same sign, then the fraction $\frac{a_1 + a_2 + a_3 + \dots + a_n}{b_1 + b_2 + b_3 + \dots + b_n}$ lies in magnitude between the greatest and least of them.
- If $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} = \dots = \frac{a_n}{b_n}$, then each of these ratios equals to $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} = \dots = \frac{a_n}{b_n} = \left(\frac{{}_1a_1^n + {}_2a_2^n + {}_3a_3^n + \dots + {}_na_n^n}{{}_1b_1^n + {}_2b_2^n + {}_3b_3^n + \dots + {}_nb_n^n} \right)^{1/n}$, where n is an integer.
- If a, b, c, d are in proportion. Then,
 - (1) $\frac{a}{b} = \frac{c}{d}$.
 - (2) $\frac{a}{c} = \frac{b}{d}$. (Alternendo)
 - (3) $\frac{a+b}{b} = \frac{c+d}{d}$. (Componendo)
 - (4) $\frac{a-b}{b} = \frac{c-d}{d}$. (Dividendo)
 - (5) $\frac{a+b}{a-b} = \frac{c+d}{c-d}$. (Componendo & Dividendo)
 - (6) $\frac{b}{a} = \frac{d}{c}$. (Invertendo)
